

# Generative AI for Everyone

Learn how generative AI works, and how to use it in your life and work



## Generative AI for Everyone

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|---------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------|
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| 👥 Instructors | Andrew Ng                                                                                                                                                       |
| 🛠️ Tools      |                                                                                                                                                                 |

## Introduction

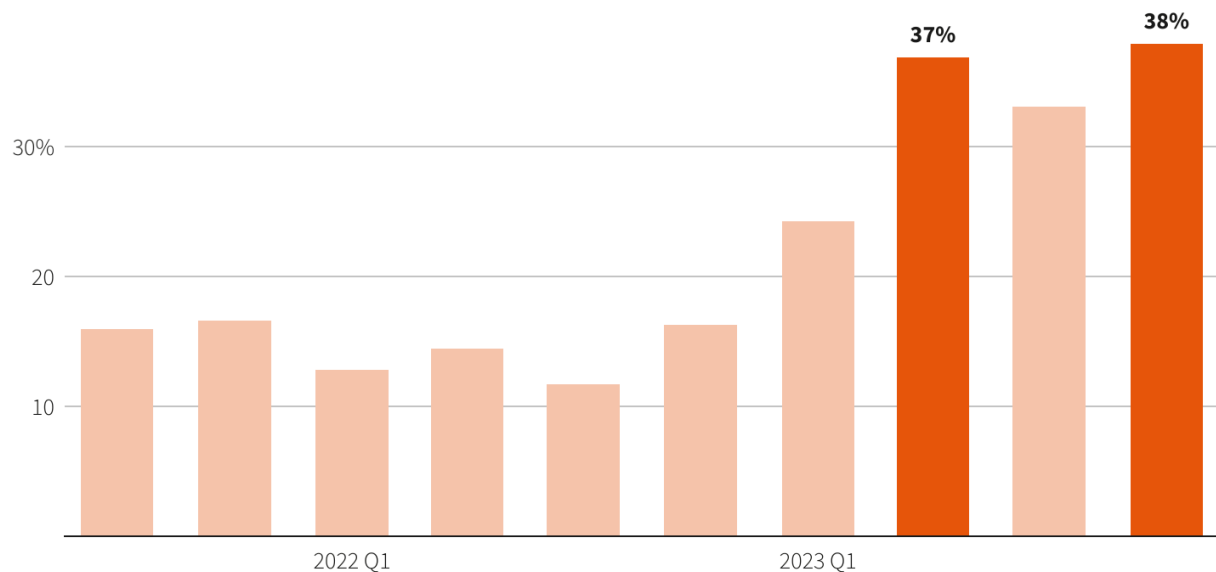
In this course

- you'll get a non-technical understanding of what Generative AI is.
  - What it can or cannot do, and how to use it in your work or business.
  - Identifying common use cases
- Generative AI projects
  - Identify and build generative AI use cases
  - Technology options
- Impact on business and society
  - How teams can take advantage of generative AI

- AI risks and responsible AI
- Does not assume any technical or AI background

## AI going strong on earnings calls

Artificial intelligence has been mentioned on 38% of S&P 500 earnings calls so far in January and February, slightly exceeding the same point during June-quarter results last year when the theme surged to prominence



Created by Thomson Reuters / Noel Randewich  
Source: LSEG

Most mentions in the tech sector followed by the energy sector in America.

<https://www.reuters.com/technology/ai-stays-front-and-center-quarterly-conference-calls-2024-01-31/>

## Generative AI could

1. Add \$2.6 - \$4.4 trillion annually to the economy
2. Raise global GDP by 7% in next 10 years
3. Impact 10% of the tasks carried out daily by 80% of the workers in US.  
Some percentage of workers can have 50% of tasks impacted by Generative AI

The Potentially Large Effects of Artificial Intelligence on Economic Growth  
(Briggs/Kodnani).

## What is Generative AI

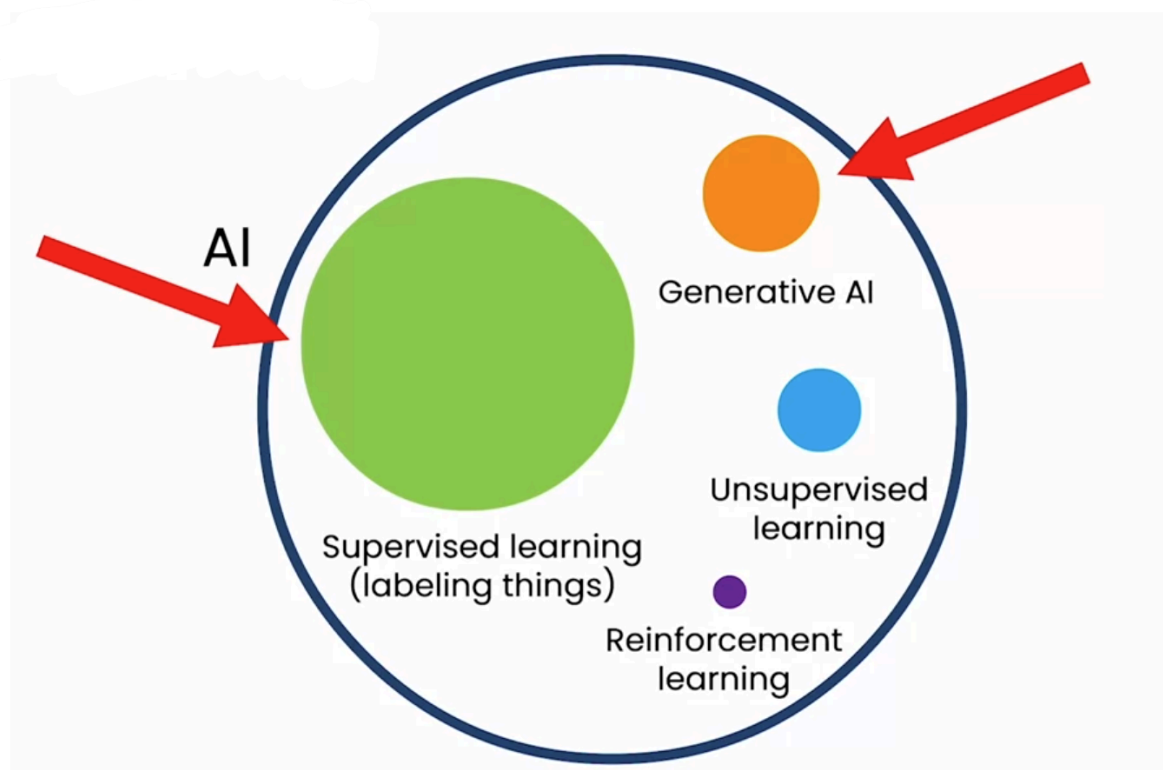
**AI systems that can produce high quality content, specifically text, images, and audio. Examples include ChatGPT, Bard, BingChat**

- Generative AI as a developer tool will be much more impactful in the long term
- AI is already pervasive - for example Google and Bing for web search, Fraud detection algorithms in Credit card payments and recommender systems in Amazon and Netflix

## How Generative AI works

### Where Generative AI fits in the AI landscape?

Think of AI as a set of tools. One of the important tools is Supervised learning, and more recent tool is generative AI. Others include unsupervised and reinforcement learning



### Supervised Learning

Given input learns the output.

| Input (A)          | Output (B)             | Application           |
|--------------------|------------------------|-----------------------|
| Email              | Spam? (0/1)            | Spam filtering        |
| Ad, user info      | Click? (0/1)           | Online advertising    |
| Image, radar info  | Position of other cars | Self-driving car      |
| X-ray image        | Diagnosis              | Healthcare            |
| Image of phone     | Defect? (0/1)          | Visual inspection     |
| Audio recording    | Text transcript        | Speech recognition    |
| Restaurant reviews | Sentiment (pos/neg)    | Reputation monitoring |

A decade of large scale supervised learning which laid the foundation for Generative AI. Initially, supervised learning involved building small models over large data. Adding more data led to performance stagnation. What researchers realized over this decade that large models can improve performance on tasks with more data.

## Generating text using 🧠 Large Language Models

LLMs are built by using supervised learning ( $A \rightarrow B$ ) to repeatedly predict the next word. For example

| My favorite food is a bagel with cream cheese |            |
|-----------------------------------------------|------------|
| Input (A)                                     | Output (B) |
| My favorite food is a                         | bagel      |
| My favorite food is a bagel                   | with       |
| My favorite food is a bagel with              | cream      |
| My favorite food is a bagel with cream        | cheese     |

When training a very large AI system with hundreds of billions of words, we get a large Language Models like ChatGPT

## LLM as a thought partner

## LLMs are giving new ways to find information

- Like what is the capital of South Africa. Sometimes, LLMs can make facts up called **Hallucination**. Can also involve back and forth to give more context.
- LLMs can also be used as a writing partner - like rewriting text or creating short fun stories

## When to use web search and when LLMs?

- Use trustworthy websites for information on healthcare related issues
- Use web pages where there is a lot of information available. Can use LLMs where there is less information available on web-searches

## AI is a general purpose technology

What is AI useful for? This is a question hard to answer because it is a general purpose technology like electricity or internet. Similar to asking what is electricity/internet good for.

Generative AI is useful for

- Brainstorming partner for writing
- Reading task - Long text to short output - Sorting emails
- Special purpose chatbots for delivery or food ordering etc.

It is useful to distinguish between two different types of LLM applications

1. **Web apps** - brainstorming and chatbot like ChatGPT, Bing Chat
2. **LLM based software** - identifying text in company emails

## Generative AI Applications

### Writing

- Brainstorming product names
- Brainstorm ideas for increasing sales for eg. cookies
- Write a press release
- Translation

Across writing we can give more context to generate specific output catering to the specifications using the context.

## Reading

Takes in a large input prompt and generates a short output

1. Proof Reading
2. Summarizing an article
3. Summarizing call center conversations
4. Customer email analysis - email routing
5. Reputation monitoring - tracking customer sentiments over time

## Chatting

- Specialized Chatbots - good at answering questions for one thing - for example travel, career coaching, cooking meals, IT service chatbots
- Chatbot customer service design points on a spectrum from humans only to chatbots only model
  - Bots supporting humans (**human-in-the-loop**)
  - Bot triages messages. For example, 10% of chats are refund requests, can be redirected while reps dealing with difficult queries
- Advice for deploying chatbots
  - Start with an internal-facing chat bots to avoid public mistakes and staff to assess the behavior of the chatbot
  - deploy with human in the loop to check for mistakes. Only if deemed safe, should be allowed to talk to customers directly.

## What LLMs can and cannot do?

**When prompting an LLM assume that you are asking a fresh college grad having lots of general knowledge to do the task for you by following some instructions. Also assume that**

- **they don't have access to internet or other resources**
- **no training specific to your company or business**

- **No memory of previous tasks completed - was not trained overtime. So you get a different fresh college graduate every time**

## Specific Limitations

### Knowledge Cutoffs

A Large language model's knowledge is frozen in time during training, so it won't have details on most recent events

### Hallucinations

For example: Give me three quotes that Shakespeare wrote about Beyonce. In this prompt, the LLM will hallucinate or make things up which can mislead people to think that this is real.

### Technical Limitation

1. Input length
  - a. LLMs accepts input prompts of certain length - called context length
  - b. Context Length - Input + Output length
2. Do not currently work with structured data
  - a. **Supervised learning would be a better technique**
3. Works best with unstructured data - images, text, audio, video etc.

### Bias and Toxicity

A LLM is trained on internet data which can have biases, so the model is likely to learn those biases as well

## Tips for Prompting

3 main tips

### 1. Be detailed and specific

and you are more likely to get a result that you want

- Give sufficient context for LLM to complete the task
- Describe the task in detail

For example

### **Initial Prompt**

*Help me write an email asking to be assigned to the legal documents project*

### **With Context**

*Help me write an email asking to be assigned to the legal documents project. I'm applying for a job on the legal documents project, which will check legal documents using LLMs. I have ample experience prompting LLMs to generate accurate text in a professional tone.*

### **With Context and Detail**

*Help me write an email asking to be assigned to the legal documents project. I'm applying for a job on the legal documents project, which will check legal documents using LLMs. I have ample experience prompting LLMs to generate accurate text in a professional tone.*

*Write a paragraph of text explaining why my background makes me a strong candidate to this project and advocate for my candidacy.*

## **2. Guide the model to think through its answer**

Add steps to the prompt for the LLM to know how to achieve the answer you want

### **Initial Prompt:**

*Brainstorm 5 names for a new cat toy*

### **Final Prompt:**

*Brainstorm 5 names for a new cat toy*

*Step 1: Come up with 5 fun, joyful words that relate to cats.*

*Step 2: For each word, come up with a rhyming name for a toy.*

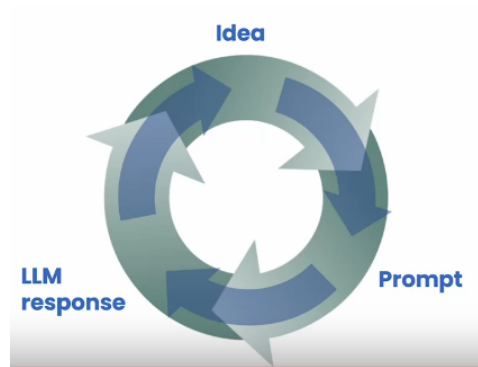
*Step 3: For each toy name, add a fun, relevant emoji.*

## **3. Experiment and iterate**

There are no perfect prompts. Adjust and improve iteratively to understand until the results are satisfactory

- Don't overthink the initial prompt
- Be careful with confidential information
- Double check before completely trusting the LLM output





## Image Generation (optional)

Image generation is mostly done via Diffusion Models .

### Training

At the heart of diffusion models is Supervised Learning. The idea is to take an input prompt and a clean image and add noise to the image over subsequent steps. The model will take the input prompt along with the noisy image to generate a less noisy image

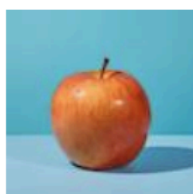


Image 1,  
"red apple"



Image 2



Image 3



Image 4

Input (A)  
Noisy image  
and caption

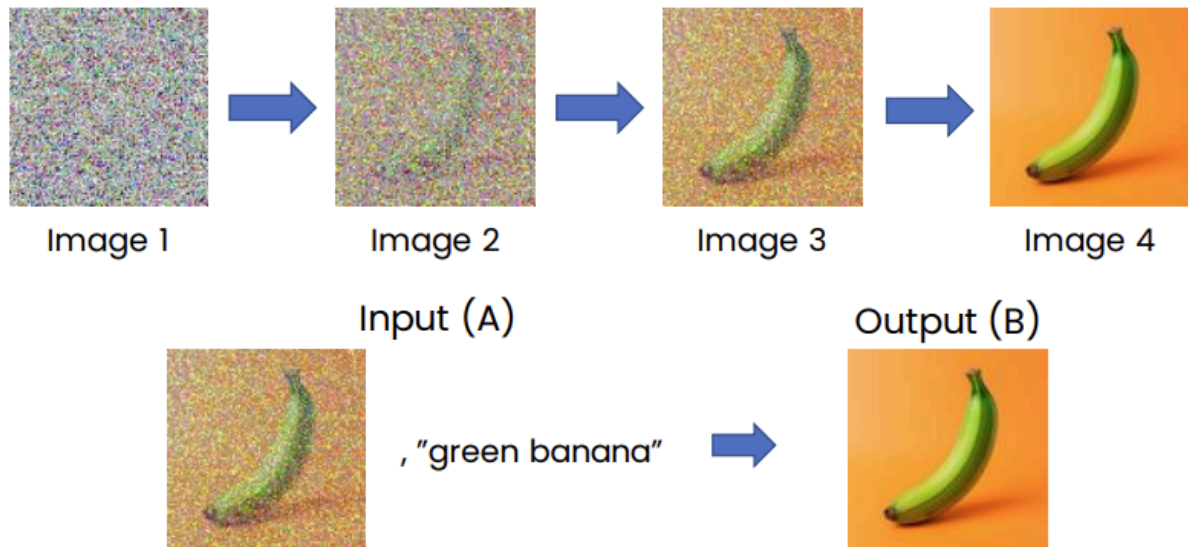


Output (B)  
Slightly  
less noisy  
image

| Input (A)            | Output (B) |
|----------------------|------------|
| Image 2, "red apple" | Image 1    |
| Image 3, "red apple" | Image 2    |
| Image 4, "red apple" | Image 3    |

### Generating Image

Image generation works exactly reverse. Instead of starting off with clean image, generation starts with random noise and a prompt. The output of first pass (image 2) is fed again with the prompt to the model to generate less noisy image (image 3) and finally image 4 the cleanest image.



Typically, while training and generating images, diffusion models undergo ~100 steps of noising and denoising before generating the final image

# Week Two

## Generative AI applications

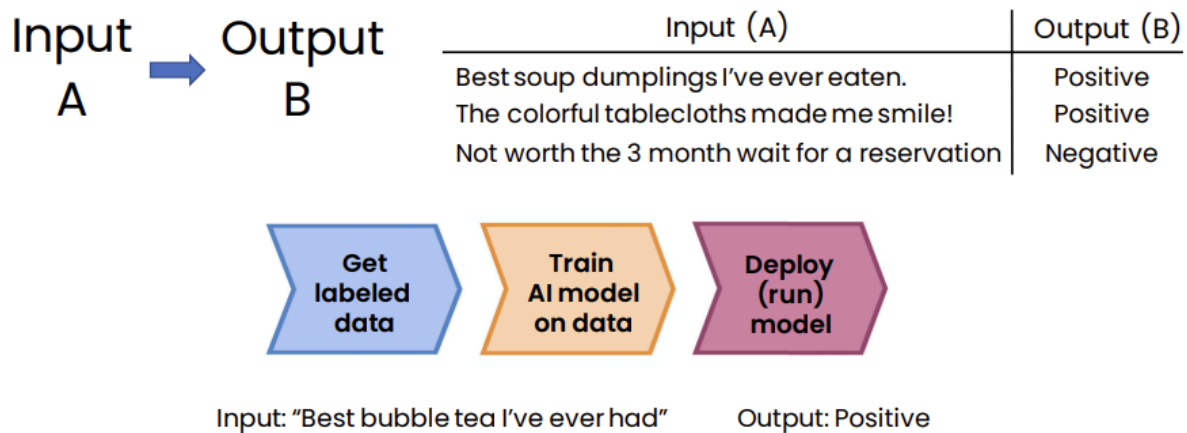
### Using generative AI in software applications

Much before generative AI, applications like reading, writing and chatbots existed where development would take months and require machine learning engineers and software developers. However, with generative AI building these applications is much easier now. For example

#### Building a reputation monitoring system

1. Get labelled data for reviews and sentiment (1 month)
2. Train a model (3 months)
3. Deploy using cloud based service (3 months)

→ development time: Months



## Prompt Based development

Can deploy the same within hours/days with prompting

```
prompt = """
    Classify the following review
    as having either a positive or
    negative sentiment:

    The banana pudding was really
    tasty!
    """

response = llm_response(prompt)
print(response)
```

Instruction text

Review text

Code to call LLM

Code to print output

## Trying generative AI code yourself (optional)

```
import openai
import os
```

```

openai.api_key = os.getenv("OPENAI_API_KEY")

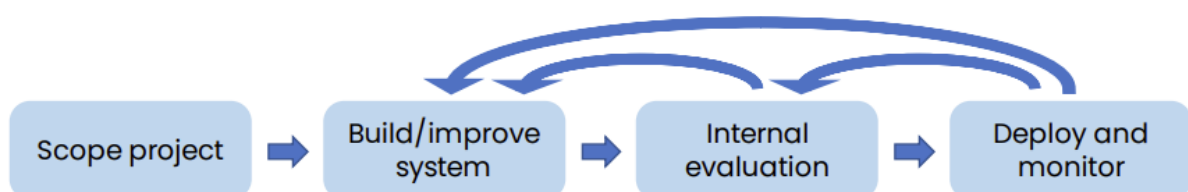
def llm_response(prompt):
    response = openai.ChatCompletion.create(
        model = 'gpt-3.5-turbo',
        messages = [{'role': 'user', 'content': prompt}],
        temperature = 0
    )
    return response.choices[0].message['content']

prompt = """
    Classify the following review as having either positive
    or negative sentiment:
    The banana pudding was really tasty
    """
response = llm_response(prompt)
print(response)

```

## Lifecycle of a Generative AI project

Building a generative AI application is a highly empirical process



1. Scope the project - decide what the software can do
2. Build/improve prototype - initial prototype in 1-2 days
3. Internal evaluation - Catch counterintuitive use cases and improve prototype
4. Deploy and monitor - Understand cases and go back to step 2 or 3

## Tools to improve performance

- Retrieval augmented generation (RAG) - Gives LLMs access to external data sources

- Fine-tune models - adapt LLMs to your specific task
- Pretrain models - train LLMs from scratch

## Cost Intuition

### How much prompting cost?

- Pricing is based on number of input and output tokens.
- Output tokens are generally more expensive.
- A token is the word used by the LLM to describe words.
  - Typically 1 token would be 3 quarters of a word. So if you generate 300 words, it would cost you about 400 tokens.
  - Very common words like "the, a, and" will be 1 token but more bigger words can be divided into multiple tokens e.g. "programming" divided into 2 tokens "program" and "ming"

### Costing by different companies

|        | OpenAI/GPT3.5      | OpenAI/GPT4      | Google/PaLM 2           | Amazon/Titan Lite  |
|--------|--------------------|------------------|-------------------------|--------------------|
| Input  | \$0.0015/1K tokens | \$0.03/1K tokens | \$0.00025/1K characters | \$0.0003/1K tokens |
| Output | \$0.002/1K tokens  | \$0.06/1K tokens | \$0.0005/1K characters  | \$0.0004/1K tokens |

### Estimating Cost

*How much would it cost to keep an adult occupied for 1 hour (Typical adult reading speed = 250 words/minute)*

- Number of Output words required =  $250 \times 60 = 15,000$
- Prompting the LLM would consume some words (assume same number as output) = 15,000
- Total words =  $15,000 + 15,000 = 30,000$
- Tokens to be consumed =  $30,000 \times 4/3 = 40,000$
- Cost per 1000 tokens = \$0.002
- Total Cost =  $\$0.002 \times 40 = \$0.08$  or 8 cents (inexpensive)

# **Retrieval augmented generation (RAG).**

RAG is a technique to allow the LLM to use specific information to generate an output more relevant to your query. For example

**Prompt: Is there a parking space of employees?**

A general purpose LLM will expect more information, but with RAG, you are also passing a document(s) (benefits, leave policy, facilities, payroll) which LLM can refer to help you answer this question

This involves the following steps

1. Given question, search relevant documents to answer
2. Incorporate retrieved text into an updated prompt
3. Generate answer from the new prompt with additional context

## **Examples or RAG applications**

1. Chat with PDF files - PDF.ai, LightPDG, Tactic, pandachat, ChatPDF, AskYourPDF, docAnalyzer.AI
2. Answer questions based on a website's articles - CourseraCoach, Snapchat, Hubspot chatbot
3. New forms of web search - BingChat, You.com a search engine based on chat like usecase

## **Think of LLMs as a reasoning engine**

- LLMs have a lot of GK, but they don't know everything
- By providing relevant context in the prompt, we can ask LLM to read a piece of text and process it to get a relevant answer
- Using it as a reasoning engine rather than as a source of information

## **Fine Tuning**

- Another way to give more information to the LLM, especially in cases where the context is bigger than the input context length
- Also this is useful when LLM needs to gain some specific language - for example, medical notes, or legal or financial documents

**Medical Notes:** Pt c/o SOB, DOE. PE: RRR, JVD absent, CTAB.

EKG: NSR. Tx: F/u w/ PCP, STAT CXR, cont. PRN O2.

**Legal:** Licensor grants to Licensee, per Section 2(a)(iii), a non-exclusive right to use the intellectual property, contingent upon compliance with fiduciary duties outlined in Section 8, paragraphs 1-4, and payment as specified in Schedule B, within 15 days hereof

- Finetuning can allow a smaller model to perform the same task compared to larger models to enable lower cost/latency to deploy or to run on mobile/laptop (edge devices)

**Both RAG and fine-tuning are cheaper to implement**

## Pre-training an LLM

- Expensive to train a large language model from scratch
- When in doubt → don't do it. Should be an option of last resort
- Requires \$10M+, many months, huge amount of data
- BloombergGPT created using 50B+ parameters by training on proprietary Bloomberg data, performs better than open source GPTs in processing financial text

## Model Size

|                  |                                                        |                             |
|------------------|--------------------------------------------------------|-----------------------------|
| 1B parameters    | pattern matching and basic knowledge of the world      | restaurant review sentiment |
| 10B parameters   | Greater world knowledge, can follow basic instructions | Food order chatbot          |
| 100B+ parameters | Rich world knowledge. Complex reasoning                | Brainstorming partner       |

## Closed or open source?

### Closed sources

- Closed source accessible by cloud interface. Easy to incorporate them in software applications
- More powerful/large models

- relatively inexpensive as they are contributed by large companies
- Some risk of vendor lock-in. Switching becomes difficult - rerunning all prompts

## Open Source

- Full control of the model
- Can run on own devices or on-premises
- Full control of data/privacy/access

## Instruction Fine-Tuning (optional)

 Large Language Models are designed to predict the next word. So if you prompt the LLM with "What is the capital of France?", the LLM may respond with "What is the capital of Germany?" as these kind of questions are prevalent in questionnaire.

To allow the LLMs to follow instruction, the LLMs are Fine Tuned on a series of good question and answer pairs in the following way

Fine-tuning

Help me brainstorm some fun museums to visit in Bogotá.

Sure, here are some suggestions:  
[...]

**Input (A)**

Help me brainstorm some fun museums to visit in Bogotá.  
Help me brainstorm some fun museums to visit in Bogotá. Sure,  
Help me brainstorm some fun museums to visit in Bogotá. Sure, here  
Help me brainstorm some fun museums to visit in Bogotá. Sure, here are  
Help me brainstorm some fun museums to visit in Bogotá. Sure, here are some

...

**Output (B)**

Sure,  
here  
are  
some  
suggestions:

...

## Reinforcement Learning with Human Feedback (RLHF)

- is a technique that can improve the quality of answers better than Instruction Fine-Tuning.
- Companies train the LLMs to be Helpful, Honest and Harmless (3Hs) - RLHF tries to accomplish this



- Steps
  - Step1: Train an answer quality (reward) model using supervised learning

| Input (A)                                              | Output (B)     |
|--------------------------------------------------------|----------------|
| Response                                               | Score (reward) |
| I'm happy to help! Here are some steps to follow [...] | 5              |
| Just try your best!                                    | 3              |
| It's hopeless – why bother?                            | 1              |

- Step 2: Have LLM generate a lot of answers. Further tune the LLM to generate more responses that get higher scores.

## Tool use and agents (optional)

### Tools

Given a prompt - *Send me a burger*, the LLM cannot just respond with *Okay! it is on the way!*. It needs to generate the order as well. So a tool, would use the prompt and call a function to generate the order and then the output is passed to the user

At the same time, to check for mission critical information, ask the user to confirm the output of the prompt whether it was expected or not.

LLMs are also bad at precise math. They can generate plausible but incorrect answers. A calculator tool can be used by the LLM to conduct precise calculations and the answer is fed back to the output

### Agents

- Use LLMs to carry out complex sequence of actions
- Cutting edge area of AI research

# WEEK3

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## Generative AI and business

### Day to Day usage of Web UIs

1. Writing Assistant and thought partner
2. Marketer for brainstorming ideas - email campaigns
3. Recruiting - Summarizing review of job candidate in 50 words or less
4. Programming - Python code

### Task Analysis of Jobs

Framework to analyzing the work task for automation. Developed by economists to identify global impacts. Same can be used by businesses to identify opportunities to use generative AI

### Automation opportunities

- AI doesn't automate jobs. It automates tasks
- Most jobs involve collection of many tasks
  - Example: Customer service rep
  - List down the tasks. Against each task, identify generative AI potential

### Augmentation or Automation

Augmentation

- Help humans with task - like dealing with rep supported by AI

Automation

- Automatically perform the task once the trust on the system is established

### How to identify potential?

The potential for augmenting and automating depends on

| Technical Feasibility<br><i>Can AI do it? How costly it is?</i>                                                                                             | Business Value<br><i>How valuable is it for AI to augment or automate this task?</i>             |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------|
| A way to think about it as a fresh college graduate following the instructions in a prompt to complete the task                                             | How much time is spent on this task?                                                             |
| If unsure, try prompting an LLM to see if you can get it to do it                                                                                           | Does doing this task significantly faster, cheaper or more consistently create substantial value |
| An engineer can also help assess if advanced techniques like <u>Retrieval augmented generation (RAG)</u> or <u>Fine Tuning</u> or other techniques can help |                                                                                                  |
| Can assess feasibility of supervised learning                                                                                                               |                                                                                                  |

## Additional Job analysis examples

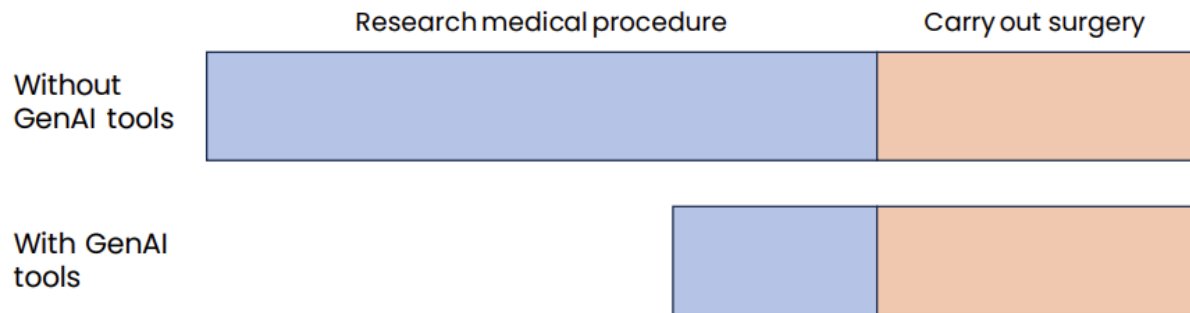
- For computer programmer, write code and documentation may be the highest potential examples subject to technical feasibility and use-cases while reviewing other's code and gathering requirements may not be
- For Lawyer, drafting and reviewing legal documents and interpreting laws, rules and regulations may have higher generative AI potential while reviewing evidence and negotiating settlements of disputes
- For landscaper, most of these tasks have low generative AI potential. So these roles may be less impacted than computer programmers and lawyers

It will be myopic just to look at generative AI potential in terms of cost savings. You can save only so much. Opportunities can be identified to improve the current workflow which can help increase revenue growth

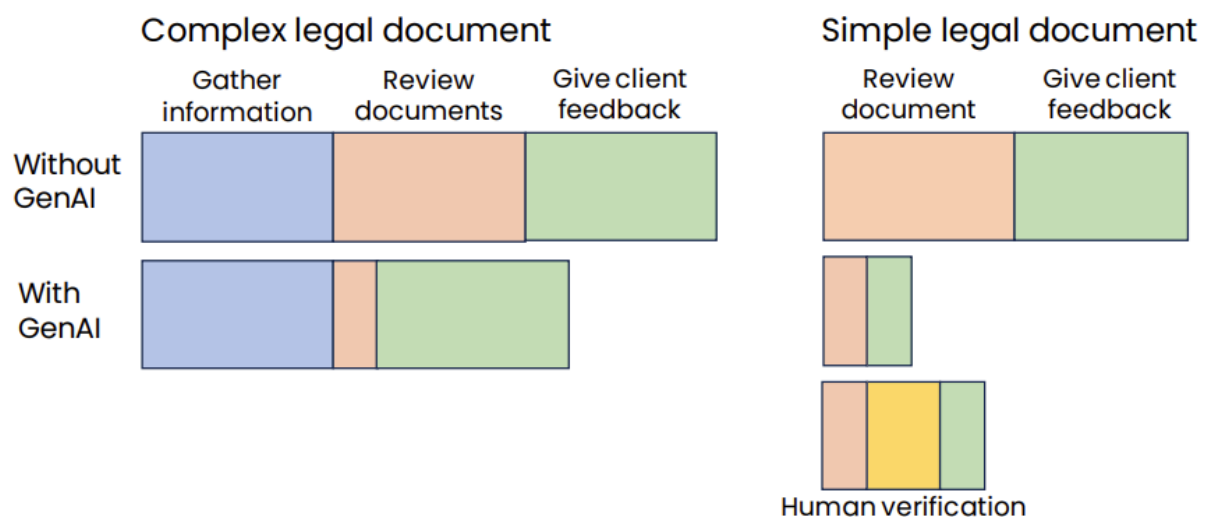
## New workflows and new opportunities

The workflow may change based on the use of GenAI tools.

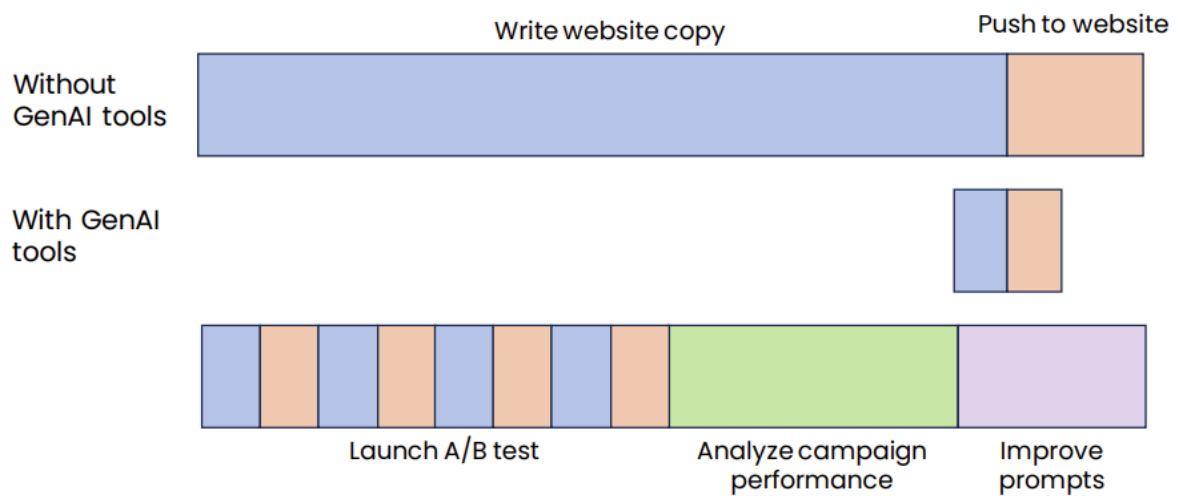
### Surgeon



## Lawyer



## Marketing Automation



## Analyzing customers' tasks

## Example: Helping customers build websites

| Tasks                   | Generative AI potential |
|-------------------------|-------------------------|
| Select website template | Low                     |
| Write title             | High                    |
| Select images           | Medium                  |
| Write copy for homepage | Medium                  |
| Optimize copy for SEO   | High                    |
| [...]                   | [...]                   |

## Teams to build generative AI software

### Common roles

- Software engineer
  - Responsible for writing software app
  - Ideally someone who has learned that basics of LLM
- Machine learning engineer
  - Responsible for implementing AI system
  - Ideally familiar with LLMs/prompting, RAG, fine-tuning
- Product Manager
  - Responsible for identifying and scoping the project
- Prompt Engineer?
  - Usually not hired as dedicated role. Prompt engineer requires additional knowledge, typically machine learning engineer.

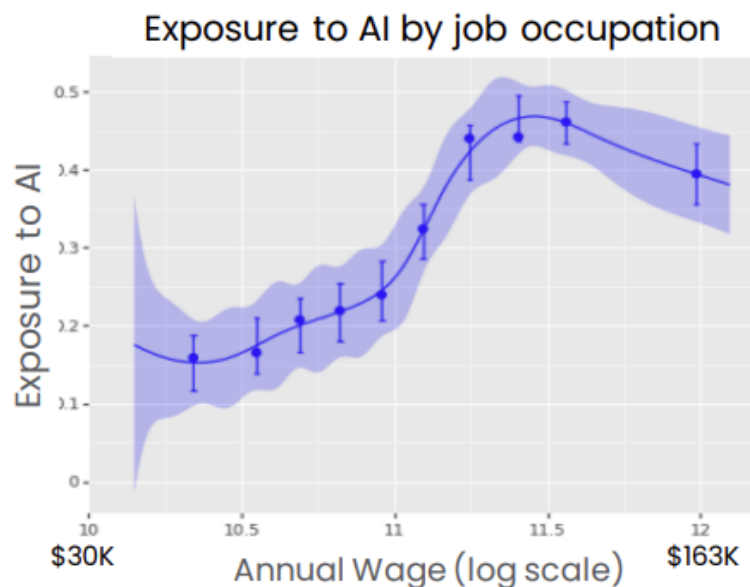
### Small team

- One person team
  - Software engineer (with some prompting know how) or
  - Machine learning engineer
- Two person team

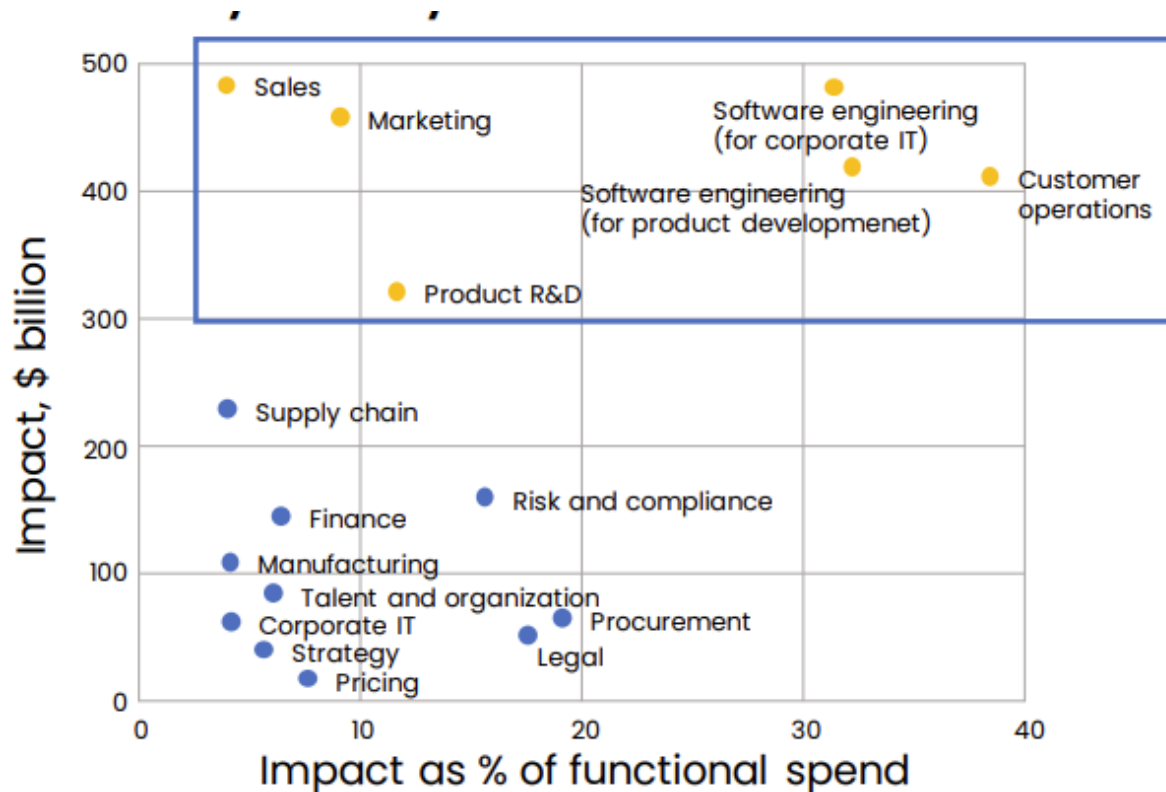
- Machine learning engineer + software engineer
- Many other configurations can also work
- Bigger teams
  - Data Engineer: Responsible for organizing data and ensuring data quality
  - Data Scientist: Responsible for analyzing data to make recommendations to guide project or business decisions
  - Project Manager: Responsible for coordinating project execution
  - Machine learning researcher: Responsible for developing and adapting advanced AI technologies to business use case

## Automation potential across sectors

- Higher paying jobs are exposing higher wage occupations to automation. Historically, low wage jobs were exposed to automation

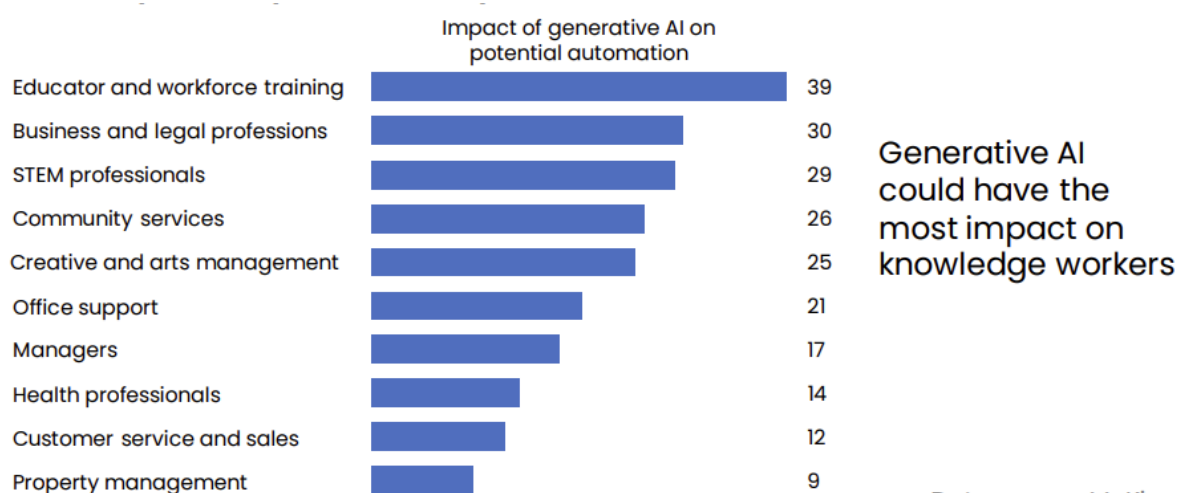


- Sales, Marketing, Software engineering, customer operations will be highly impacted



Source: McKinsey report

- Impact by Industry - these sectors were not prone to automation. Most impact on knowledge workers - expertise, critical thinking and interpersonal skills



Data source: McKinsey

## Generative AI and society

### Concerns about AI

### 1. Whether AI will amplify humanity's worst impulses and biases?

Can make them less biased by Fine Tuning or using advanced techniques such as Reinforcement Learning with Human Feedback (RLHF)

- a. RLHF to be used to train LLMs via supervised learning on unbiased reward model

### 2. Who is going to lose job?

Geff Hinton made a comment in 2016 that Deep learning is progressing so fast that in 5 years, radiologist will be replaced by AI. But it has been more than 5 years and not a single radiologist has lost their job. That is because radiologist perform more than 30 tasks, and interpreting X-rays is only one of the tasks. Also, interpreting x-rays is harder than imagined.

*AI won't replace radiologists*

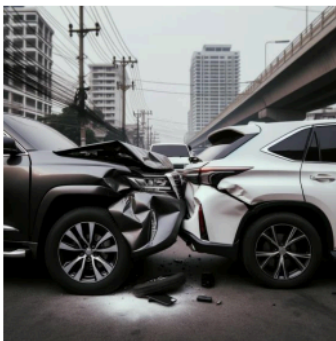
*But radiologists who use AI will replace radiologists that don't*

*- Curtis Langlotz, Professor of Radiology, Stanford University*

### 3. Human Extinction

Extinction arguments are not concrete and specific. Most arguments boil down to it could happen. and is true for any new type of technology

Self-driving car crashes



2010 Stock Market Flash crash



Unjust sentencing in criminal cases



Perfect control is not needed for it to be valuable and safe. For example, airplanes, we cannot control turbulence. On the flip side, AI will be key part to the solutions for real risks to humanity like climate change and pandemics

## Artificial General Intelligence (AGI)

General purpose technology not equal to Artificial General intelligence. AGI by definition can do any intellectual task that a human can. We can expect AGI to



- learn to drive with 20 hours of practices
- Complete PhD thesis within 5 years: Currently, it can only do some part of research
- Conduct all tasks by a computer programmer

Though AI is powerful and better than humans at some specific tasks, expecting AI to perform like humans is a very high bar.

## Responsible AI

Many governments and companies are publishing frameworks for responsible AI.

Dimensions of responsible AI

- Fairness - Ensuring AI does not perpetuate or amplify biases
- Transparency - Making AI systems and their decisions understandable to stakeholders impacted
- Privacy - Protecting user data and confidentiality
- Security - Safeguarding AI systems from malicious attacks
- Ethical Use - use only for beneficial purposes

## Tips for responsible AI

- Build a culture to debate and discuss about ethical issues
- Brainstorm how things can go wrong
- Work with a diverse team including perspective from all stakeholders

## Course Summary

- How generative AI works
  - What it can or cannot do
  - Covered common use cases
- Generative AI projects
  - Lifecycle

- Technology options - Prompting, Retrieval augmented generation (RAG), Fine Tuning
- Implications on business and society
  - Analyzing tasks in jobs for automation and augmentation potential
  - Societal concerns and responsible AI

## Building a more intelligent world

- Intelligence is power to apply knowledge and skills to make good decisions.
- Human intelligence is expensive - feeding, education and training a wise human being. That is why only the wealthiest people are able to afford and hire large amounts of intelligent people
- Artificial intelligence is cheap - AI has the potential to give every individual the ability to hire intelligence at low cost
- AI is the new electricity with the potential to revolutionize all industries and all corners of human life. The fear of AI today is similar to the fear of electricity when it was new. Back then, people were terrified of electrocution or of electricity sparking devastating fires. Today, electricity still has dangers, but I don't think any of us would give up light, heat, and refrigeration for fear of electrocution
- Looking beyond AI, the world has many problems such as climate change, pandemics, war, and many others, and many of these problems need urgent solutions. To solve them, we will need all the intelligence, including all the artificial intelligence we can master.

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